1.12 TEST OF HYPOTHESES FOR EQUALITY OF MEANS OF TWO POPULATIONS (*POPULATION VARIANCES ARE UNKNOWN*)

Procedure:

Step-1 : Let μ_X and σ_X^2 be respectively the mean and the variance of Population -1. Also, let μ_Y and σ_Y^2 be respectively the mean and the variance of Population -2 under study. Here σ_X^2 and σ_Y^2 are assumed to be unknown.

Frame the null hypothesis as H_0: $\mu_X = \mu_Y$ and choose the suitable alternative hypothesis from

(i) H_1: $\mu_X \neq \mu_Y$ (ii) H_1: $\mu_X > \mu_Y$ (iii) H_1: $\mu_X < \mu_Y$

Step 2 : Let $(X_1, X_2, \ldots, X_m)$ be a random sample of m observations drawn from Population-1 and $(Y_1, Y_2, \ldots, Y_n)$ be a random sample of n observations drawn from Population-2, where m and n are large ($m \geq 30$ and $n \geq 30$). Here, these two samples are assumed to be independent.

Step 3 : Specify the level of significance, α.

Step 4 : Consider the test statistic

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{m} + \frac{S_Y^2}{n}}} \text{ under } H_0 \text{ (i.e., } \mu_X = \mu_Y).$$

i.e., the above test statistic is obtained from Z considered in the test described in Section 1.11 by substituting S_X^2 and S_Y^2 respectively for σ_X^2 and σ_Y^2

The approximate sampling distribution of the test statistic $Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_X^2}{m} + \frac{S_Y^2}{n}}}$ under H_0 is the $N(0,1)$ distribution.

Step 5 : Calculate the value of Z for the given samples $(x_1, x_2, \ldots, x_m)$ and $(y_1, y_2, \ldots, y_n)$ as

$$z_0 = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_X^2}{m} + \frac{s_Y^2}{n}}}$$

.

Here $\bar{x}$ and $\bar{y}$ are respectively the values of $\bar{X}$ and $\bar{Y}$ for the given samples. Also, s_X^2 *and* s_Y^2 are respectively the values of S_X^2 and S_Y^2 for the given samples.

Step 6 : Find the critical value, z_e, corresponding to α and H_1 from the following table

Alternative Hypothesis (H_1)	$\mu_X \neq \mu_Y$	$\mu_X > \mu_Y$	$\mu_X < \mu_Y$
Critical Value (z_e)	$z_{\alpha/2}$	z_α	$-z_\alpha$

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